

# Enterprise Learning & Workforce Management Platform

## Milestone 2 – Core Learning Experience & Admin Foundations

*Student Dashboard · Courses & Lessons · Assessments · Certificates · Admin Dashboard*

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### 1. Objective

Milestone 2 builds the core learning loop of the platform on top of the authentication foundation from Milestone 1. Once a user logs in, they need somewhere to land, content to learn from, a way to prove what they've learned, and a credential to show for it. In parallel, the platform's administrators need the corresponding back-office tools to manage that same content and those same students.

The main objectives were:

- Build the authenticated Student Dashboard as the landing screen after login.
- Model courses as a structured hierarchy of modules and lessons.
- Implement enrollment so students can join a course and have progress tracked.
- Implement an Assessment module that measures understanding of course content.
- Auto-generate verifiable Certificates once a student passes a course assessment.
- Introduce the Admin Dashboard, gated by the ROLE\_ADMIN authority from Milestone 1.
- Give admins CRUD control over courses/modules, oversight of users, and oversight of certificates.

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### 2. Student Dashboard

The Student Dashboard is the first screen a user sees immediately after a successful login or registration — the authenticated counterpart to the public landing page from Milestone 1.

## 2.1 Purpose & Layout

The dashboard gives a student an immediate snapshot of where they stand: which courses they are enrolled in, and a shortcut into the deeper progress-analytics view that is completed in Milestone 3. It is intentionally lightweight on this screen — a summary, not a full report — so it loads quickly and orients the student before they choose what to do next.

## 2.2 Data Flow

On mount, the dashboard component calls the backend for the current student's enrollment summary. The flow is:

```

Dashboard Page Loads
↓
Axios GET /api/enrollments/my
↓
JwtAuthFilter validates token
↓
EnrollmentController → EnrollmentService
↓
Query enrollments WHERE user_id = current user
↓
Return list of {course, progress}
↓
React renders dashboard cards

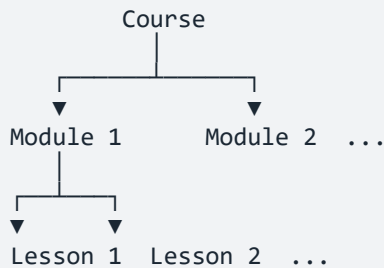
```

If the student has no enrollments yet, the dashboard renders an empty state that nudges them toward the course catalog rather than showing a blank screen.

## 3. Courses & Content Structure

Courses are the core content unit of the platform. They are modeled as a three-level hierarchy so that content can be organised and tracked at a granular level.

### 3.1 Course → Module → Lesson Hierarchy



A Course entity holds the title, level, duration, rating, price, and description. Each Course owns a set of CourseModules, and each CourseModule owns a set of Lessons. This structure lets the platform track completion at the finest level (a single lesson) and roll that information up into module-level and course-level progress.

### 3.2 Enrollment Flow

A student joins a course by enrolling in it. Enrollment creates the link between the student and the course, and is the record against which all later progress is measured.

```

Student clicks Enroll
↓
POST /api/enrollments
↓
EnrollmentController
↓
Check not already enrolled (unique user_id + course_id)
↓
Create enrollments row (progress = 0)
↓
Student can now access course content

```

### 3.3 Lesson Progress Tracking

As a student completes a lesson, the frontend calls the backend to mark it done. A LessonProgress record is written per lesson, and the course-level progress percentage on the enrollments row is recalculated from the ratio of completed lessons to total lessons in the course. This is the same progress percentage the Student Dashboard and, in Milestone 3, the My Progress analytics page both read from.

## 4. Assessment Module

The Assessment module is the checkpoint between consuming course content and being awarded a certificate. Each enrolled course unlocks exactly one assessment.

### 4.1 Assessment Unlock Flow

```

Student enrolls in Course
↓
Assessment becomes available for that course
↓
Student starts Assessment
↓
AssessmentController serves questions
↓
Student submits answers
↓
Score calculated and recorded

```

### 4.2 Scoring & Eligibility

The recorded score is the single source of truth for whether a student is eligible for a certificate. This deliberately separates 'watched the lessons' from 'demonstrated understanding' — a student cannot receive a certificate simply by marking lessons as complete without passing the corresponding assessment.

## 5. Certificate Generation & Verification

Once a student passes a course's assessment, the platform automatically issues a certificate — no manual admin action is required for the common case.

## 5.1 Auto-Generation Flow

```

Assessment passed
↓
POST /api/certificates/generate/{courseId}
↓
CertificateController → CertificateService
↓
Generate unique certificateId (e.g. SSLN-2026-XXXXXX)
↓
Snapshot student name + course name at issue time
↓
Generate QR code + verificationToken
↓
Save Certificate row
↓
Certificate available to student instantly

```

Capturing the student's name and course name as a snapshot on the certificate (rather than only referencing the live user/course rows) means the certificate stays valid and displays correctly even if the student later edits their profile name or the course title changes.

## 5.2 PDF Export

Certificates can be downloaded as a PDF using the OpenPDF library on the backend, so students have a portable file they can attach to resumes, emails, or job applications rather than only a link to the platform.

## 5.3 Public Verification

Every certificate can be independently verified without logging in, using its certificateId. This is the platform's anti-fraud mechanism — an employer or recruiter can confirm a certificate is genuine without needing an account.

```

Anyone opens verification link
↓
GET /api/certificates/verify/{certificateId}
↓
No authentication required
↓
Certificate found? → Yes → Show verified details
                  → No  → Show "not found / invalid"

```

# 6. Admin Dashboard

Milestone 2 introduces the Admin Dashboard — a separate area of the application accessible only to users with ROLE\_ADMIN, using the exact same JwtAuthFilter and role check mechanism established in Milestone 1.

## 6.1 Admin Access Control

```

Admin logs in (same AuthController as students)
↓
JWT issued with role = ADMIN
↓
Request to /api/admin/**

```

```

↓
JwtAuthFilter checks role claim
↓
role == ADMIN? → Yes → Allow
                → No  → 403 Forbidden

```

## 6.2 Course & User Management

Administrators get full CRUD control over courses, modules, and lessons — the same content hierarchy students consume — plus the ability to search all registered users and enable/disable an account via its status field, without deleting the underlying data.

## 6.3 Certificate Oversight

Admins can view every certificate issued on the platform and revoke one if necessary (for example, in case of a data error or academic-integrity issue), keeping the verification system trustworthy.

## 6.4 Stats Overview

A dedicated stats endpoint aggregates headline numbers — total users, total courses, total enrollments — for the Admin Dashboard's home screen, giving administrators an at-a-glance read on platform health before drilling into any specific module.

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## 7. Core Backend Classes

### **CourseController.java / CourseModuleController.java / LessonController.java**

Expose CRUD REST endpoints for the Course → Module → Lesson hierarchy, used by both the student-facing catalog and the admin management screens.

### **EnrollmentController.java**

Handles enrolling a student into a course, listing a student's enrollments, and updating progress as lessons are completed.

### **AssessmentController.java**

Serves assessment questions for an enrolled course and accepts submissions, recording the resulting score.

### **CertificateController.java**

Handles certificate generation, retrieval of a student's certificate library, PDF download, and public verification by certificate ID.

### **AdminController.java**

Exposes the admin-only stats overview and ties together the course, user, and certificate management endpoints under role-gated routes.

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## 8. Database Design for This Milestone

The following tables were introduced or actively used to support this milestone's features:

- courses — title, level, duration, rating, price, description

- `course_modules` — belongs to a course; groups related lessons
- `lessons` — belongs to a module; the smallest unit of content
- `lesson_progress` — tracks per-student, per-lesson completion
- `enrollments` — links a user to a course; unique on (`user_id`, `course_id`); holds the rolled-up progress percentage
- `certificates` — `certificateId`, snapshot student/course names, grade, issue/completion dates, QR code, verification token

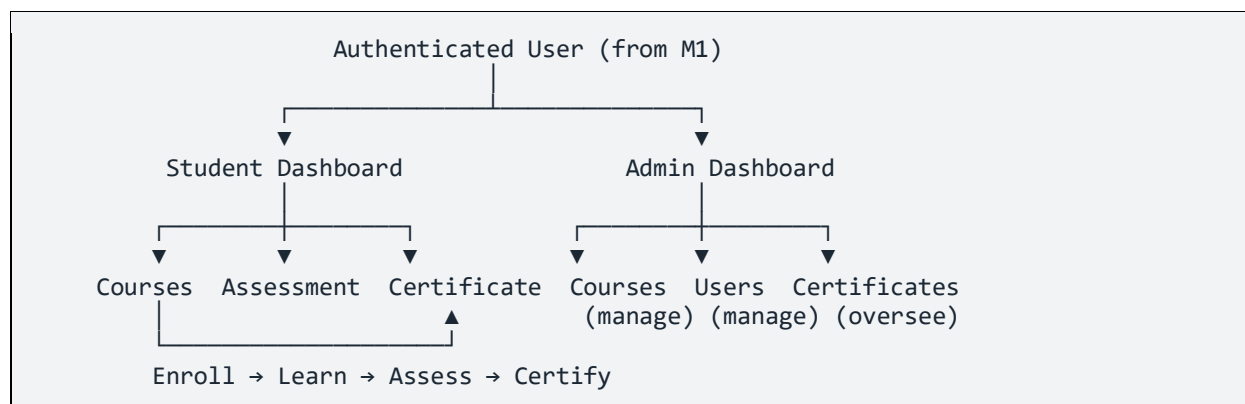
Relationships:

```
users (1) —< enrollments >— (1) courses
courses (1) —< course_modules >— (1) lessons
lessons (1) —< lesson_progress
users (1) —< certificates >— (1) courses
```

## 9. Implementation Note – Preventing Duplicate Certificates

Because certificate generation is triggered automatically the moment an assessment is passed, the `certificateId` column carries a unique constraint at the database level in addition to the application-level check that a certificate doesn't already exist for that student/course pair. This guards against a duplicate certificate being created if a student re-submits a passing assessment or the generation endpoint is called more than once.

## 10. Complete Milestone 2 Architecture



## 11. Conclusion

Milestone 2 turned the authenticated shell from Milestone 1 into a working learning platform. Students can now land on a dashboard, enroll in structured courses, progress through modules and lessons, prove their understanding through an assessment, and walk away with a verifiable certificate — while administrators gained the matching back-office tools to manage that same content and oversee the student base.

With the core learning loop and admin foundations in place, the platform is ready for Milestone 3, which builds on this foundation with deeper progress analytics, skill-building practice, competitive contests, and a structured career roadmap.